

LEARNING-BASED APPROACH FOR LICENSE PLATE RECOGNITION

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Abstract

This paper presents a learning-based approach for the construction of license plate recognition system. The system consists of three modules. They are respectively, car detection module, license plate segmentation module and recognition module. Car detection module detects a car in the given image sequence obtained from the camera with simple color-based approach. Segmentation module extracts the license plate in detected car image using neural networks (NNs) as filters for analyzing the color and texture properties of license plate. Recognition module then reads characters in detected license plate with support vector machine (SVM)-based character recognizer. The system has been tested with 1000 video sequences obtained from tollgate and parking lot, etc. and have shown the following performances on average: Car detection rate 100%, segmentation rate 97.5%, and character recognition rate about 97.2%.

1. INTRODUCTION

Automatic recognition of license plate is an essential stage in intelligent traffic system and many methods have been developed for the construction of license plate recognition (LPR) system [1-5]. This area is challenging because it

requires an integration of many of computer vision problem solvers. These vision problems include object detection and character recognition, which involve the processing of “concepts” like license plate or characters. The problem is that it is often difficult to express these concepts with algorithmic methods. In this case, example-based learning machines, such as neural networks (NNs) or support vector machines (SVMs) can be used to provide acceptable alternatives.

In this paper, we propose a learning-based approach for the construction of license plate recognition system. The system consists of three modules [6]: car detection module, license plate segmentation module and license plate recognition module, respectively. We make use of NNs as filters, which analyze color and textural properties of license plate and thus use them for the segmentation of license plate. SVMs are then used for recognizing characters in the license plate. Each of learning machines is trained with examples by supervised learning algorithm.

This paper is organized as follow. Section 2 describes the proposed system and is divided into three sub-sections. Section 2.1 specifies the car detection module that gives candidate area from detected image. Section 2.2 presents segmentation of license plate and neural network based filters that extract license plate from candidate area. Section 2.3 presents character extractor and neural network based recognizer. Section 3 makes a description of system performance, equipment, and environments. In final section we summarize our system.

2. SYSTEM DESCRIPTION

The proposed system has been designed to recognize license plate located in front of vehicles. Input of this system is an image sequence acquired by video camera. Its output is the identity of car passing in by a specified region represented as recognized characters. It is consisted of three main modules: car detection, segmentation and recognition of license plate. The structure of system is

shown in Figure 1. Each of these modules will be described in detail in the following.

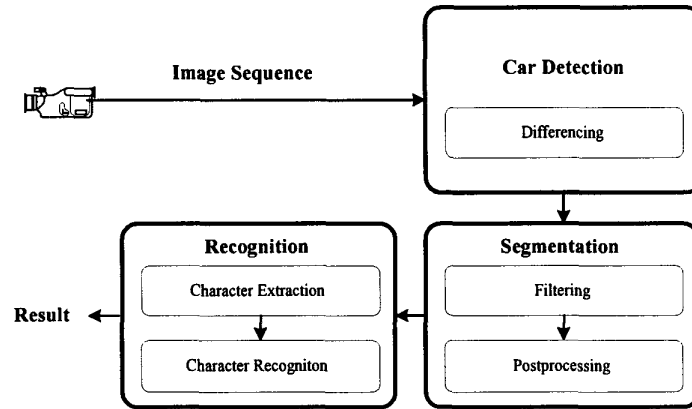


Figure1. The block structure of system

2.1 Car Detection

The car detection module detects car in given image sequence. Input of module is sequence of color image acquired by camera and its output is a candidate area image that captures a car passing by specified region. Candidate area is a region of image where there is expected to be license plate. It is reason of using candidate area to reduce total computing time. As the system is designed to pass a fixed region, we have previously a rough knowledge of location. The method of detecting car uses the rate of color change in vertical sensor. Vertical sensor is a special region of image. The position of vertical sensor provides the information of license plate location. The difference between vertical sensor of continuous images show whether there is a car or not. If difference value is higher than threshold value, it is regarded that the current image contains car. We get threshold value by experiment.

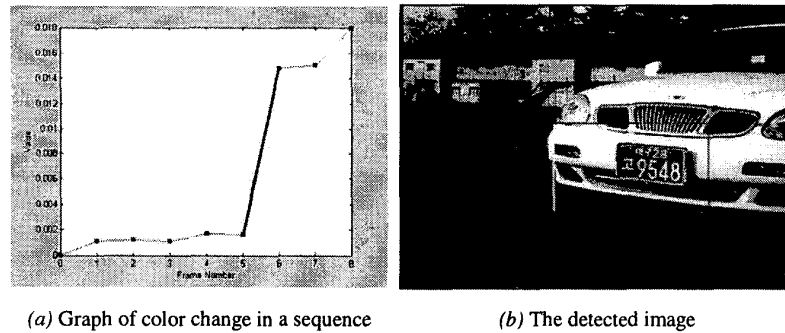


Figure 2. Example of car detection

Figure 2 (a) indicates the graph of changing color on vertical sensor in a sequence. When 6th frame is examined, the car is detected Figure 2 (b). Vertical white line in (b) indicates vertical sensor. Rectangle consisted of red line and vertical sensor indicates candidate area. If we have 3 frame/second input from input device, there is no problem to detect license plate in candidate area.

2.2 Segmentation of License Plate

To find the license plate from the candidate area, this module first applies two neural network-based filters to an input and then it uses a post-processor to combine the two filtered images in order to locate license plates. These two neural networks called vertical and horizontal filters respectively, examine small windows of vertical and horizontal cross-sections of an image and decide whether each window contains a license plate. Cross-sections contain sufficient information for distinguishing a plate from the background[5].

We use two TDNNs as horizontal and vertical filters. A TDNN is a multi-layer feed-forward network whose hidden neurons and output neurons are replicated across a time. A TDNN has the ability to represent relationships between events in time and to use them in trading relations for making optimal decisions [7]. Although a static image does not have an explicit time direction, the

TDNN is used as if the cross sections of image are stretched to a time direction [5].

Parameters	Horizontal TDNN	Vertical TDNN
Input window len./wid.	50/3	30/3
Num. of sub-sample. Steps	3/3/1	4/2/1
Rlen/Rwid of first layer	10/5	6/5
Rlen/Rwid of second layer	5/5	6/5
Rlen/Rwid of third layer	9/1	6/1
Num. Of indep. Weights	320	270

Table 1. TDNN specifications

The network specifications are shown in Table 1. The length and the width of receptive field of each layer are marked with Rlen and Rwid. Figure 3 shows the overall structure of a TDNN. Both TDNNs are trained with back-propagation algorithm. We train these networks on samples from the pure plates and backgrounds, excluding the boundaries between plate and background.

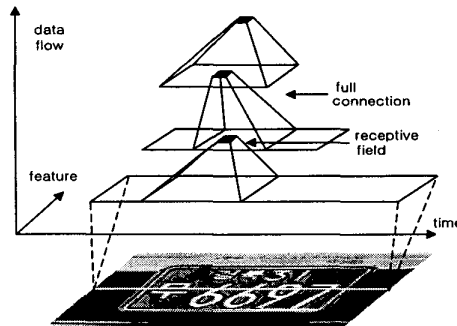


Figure 3. Structure of the TDNN (time-delay neural network).

The post-processor combines two filtered images and locates bounding boxes of license plates in the image. The post-processing consists of three steps. First, plate candidate regions are obtained from two filtered images. In

the horizontally filtered image, the horizontal neighbor pixels of pixels those have an intensity value larger than 0 are marked as horizontal candidate pixels. The vertical candidate pixels are similarly obtained. Intersections of these horizontal and vertical candidate pixels indicate the plate candidate regions. The second step removes any small noisy elements and merges neighboring candidate regions. The final step places bounding boxes around the plate candidate regions using license plate characteristics. Size, shape and height/width-ratio are all used as license plate characteristics [5]. Figure 4 shows an example of a license plate segmentation.

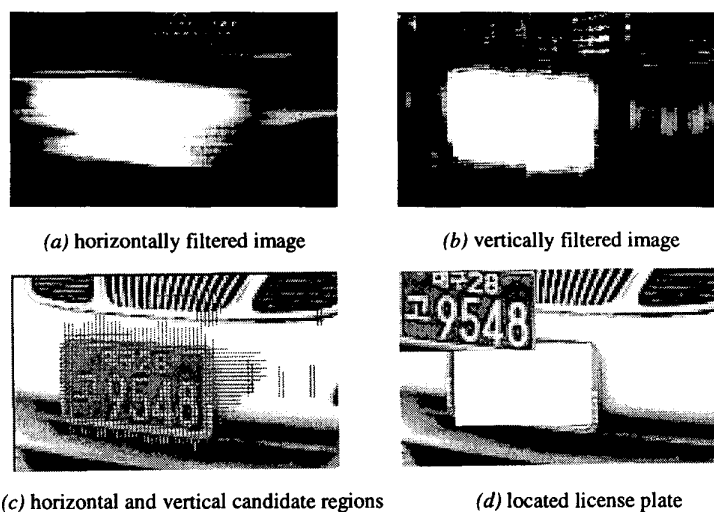


Figure 4 Example of license plate location

2.3 Recognition of License Plate

In order to get the identity of a vehicle from the license plate, recognition of the license plate module extracts and recognizes characters. For Korean license plate, there are two rows of characters in a plate. Upper row contains two Korean characters and one or two numerals. Lower row contains one Korean character

and four numerals. Two characters in the first row represent the area code. Because the total number of area code is 15, two characters are recognized simultaneously. Lower character represents the usage of vehicle. The number of total class for lower character is 30.

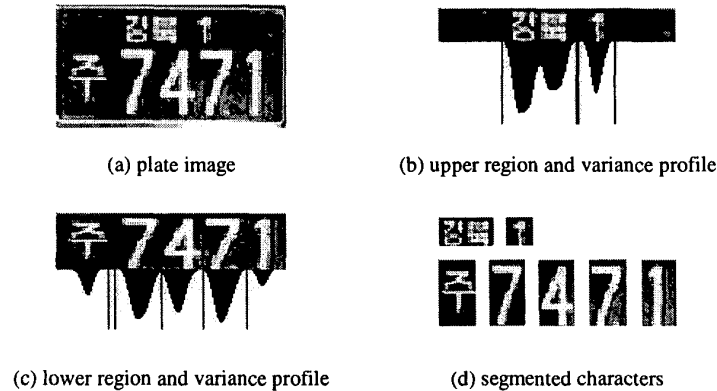


Figure 5. An example of segmentation process

Thus Recognition of license plate module first vertically segments a plate into two regions and horizontally segments each region. Because of low visual quality, the threshold method does not guarantee good extraction of characters. Intensity level profile analysis yields the segmentation. However, instead of simply using projection profile alone, we also use the variance of gray levels. Thus we use the linear sum of intensity projection and intensity variance of projection direction. This enables recognition of license plate module do not sensitive to the change of illumination. Figure 5 Shows example of extraction process: (a) plate image which is input of recognition of license module, (b) is upper row region and intensity-variance profile (c) is lower row region and intensity-variance profile (both profiles are smoothed), (d) is segmented characters.

Each extracted character image is firstly histogram equalized and uniformly divided into 8×8 regions yielding 64 dimensional mesh vector. Four SVMs-based character recognizers are used to recognize the characters. Each recognizers

indexed by SVM_{uc} , SVM_{un} , SVM_{lc} , SVM_{ln} , respectively, recognizes upper characters, upper numerals, lower character and lower numerals. They use one-per-class decomposition method to classify between multi-classes characters. In case of number, we adopt 10 SVMs respectively, 0~9, and select the maximum value of their outputs.

3. SYSTEM PERFORMANCE

In order to evaluate the performance of proposed system, we have performed each module in system with 1000 video clips, each of which is consisted of about 10 frames and contains a vehicle passing through a specified region including the tollgate, parking lot, etc. They are consisted of 600 training set and 400 testing set. The sizes of frames in video clips are 640×480 and the sizes of the license plates in frames, which are captured by input device, camera and capture board, range from 82×52 to 134×52 with 24bit/pixel.

To analyze the performance of car detection module, we utilize video clips for input. The output is candidate area where is a part of input image. The size of candidate area is 250×150. We decide whether it is correctly detected or not by using whether it is included or not in candidate area. If entire license plate does not include in here, it is taken for an error. The rate of correct detection is 100%.

To test the performance of license plate segmentation module, we should define the criterion for the correct location. We determine whether the result is correct or not by comparing the plate location by segmentation of license plate module with that of manually obtained. Performance criteria can be found in [8]. The rate for correct license plate rate with segmentation of license plate is 97.5%. The missed plates are reflecting sunlight, thus difficult to locate. The falsely located regions are intersected with real license plates yet did not fit the criteria for a correct location.

SVMs in license plate recognition module have been trained with training

algorithm. The average character recognition rate with recognition of license plate module on four types of character is 97.2%. Recognition results for each character category is shown in Table 3: Each of *UC*, *UN*, *LC*, *LN* represents the upper character, upper numeral, lower character, lower numeral respectively.

	<i>UC</i>	<i>UN</i>	<i>LC</i>	<i>LN</i>
Average Recognition rate (%)	98.3	97.2	95.4	98.0

Table 3. Performance of character recognizer

Total license plate recognition is thus 94.7%. Table 4 summarizes the results.

Module	Detection	Segmentation	Recognition	Total
Performance (%)	100	97.5	97.2	94.7

Table 4. License plate recognition results

The system has been implemented on a Pentium II PC. Program modules are written in Visual C++6.0 language in a Windows environment. More than half of CPU time is spent in segmentation of license plate module for TDNN-based filtering. The time spent to run the segmentation and recognition of license plate is about 1 seconds per image. As the operation of car detection module is very simple, the processing time of it is negligible.

4. CONCLUSION

In this paper we have presented a vehicle license plate recognition system. It consists of three modules, each of which was an integration of various vision technologies. The system was tested on a set of about 1000 video clips. The obtained results showed that the system was able to work efficiently in practice. It

was possible to replace each module by other delicate module to improve the overall performance of system.

Acknowledgement

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